Exp. no 13 STUDY OF ... CABLES

Aim:-

Study of different types of network cable

Different type of cables used in networking are

1. Unshielded twisted pair (UTP) cable.
2. shielded Twisted pair (STP) cable
3. Co axial Cable
4. Fibre optic cable.

Cable type	Category	Maximum Data Transmission	Advantage / Disadvantage	Applica / use
UTP	3	10 bps	Advantage • cheaper in cost	• 10 Base-T ethernet
	5	upto 100 mbps	• easy to install	• Fast ethernet, gigabit ethernet
	5e	1 gbps	Disadvantage • More prone to EMI and noise	
STP	6, 6a	10 gbps	Advantage • shield • Faster than UTP • Less susceptible to noise	• GB ethern 10G ethern (55 m) wide used in data cent
SSTP	7	10 Gbps	Disadvantage • Expensive • Greater installation effort	• GB ethernet 10G ether (100 m)

Co-axial Cable	RG-6 RG-59 RG-11	10-100 Mbps	• High bandwidth • Immune to interference • Low loss bandwidth • Versatile Disadvantage: • Limited distance • Cost • Size is bulky	• Speed of signal is 500m taking network High speed internet connections.
Fibre optics cable	Single mode multi mode	100 Gbps	Advantage: • High speed • High bandwidth • High security • Long distance Disadvantage:- • Expensive • Requires skills installers	• Maximum distance of fibre optics cable is around 100 meters.

The T568 standard color code sequence or RJ45 Pinout Scheme:-

To wire an RJ45 connector for an ethernet cable follow either T568A or T568B standard.

Holding the connector with gold pin facing up and plastic clip away from us.

The most common standard, T568B, uses this exact wire order.

1. White / Orange
2. Orange
3. White / Green
4. Blue
5. White / Blue
6. Green
7. White / Brown
8. Brown.

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